



Master App Guide

Complete Application Reference

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This document supersedes all previous Master App Guide versions.

Includes: Self-Hosting Guide references, HA Flows A-I, and all recent UI changes.

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1. Project Overview & Architecture

UniCal is a full-stack academic task management app for TMU students. A unified dashboard for coursework, schedules, media, home automation, and more.

Technology Stack

- Frontend: React 18 + TypeScript + Vite + TailwindCSS + shadcn/ui
- Backend: Express.js + TypeScript
- Database: PostgreSQL via Drizzle ORM
- Hosting: Replit (dev) -> Raspberry Pi 5 (production)
- Dev server: npm run dev (Express + Vite on port 5000)

Dashboard Features

- Weekly calendar grid with tasks, events, time blocks
- Weather widget (Open-Meteo, free)
- D2L ticker + news ticker
- Homework panel with full task management
- Spotify player, Alexa/HA speaker controls
- PDF reader with TTS (Cat Washroom System)
- CRCU shift calendar, morning review, monthly reports
- OneDrive sync, contacts, code checker with AI

2. Three-Tier Authentication

Tier 1 - Code 5747 (Full Access)

All features: dashboard, homework, calendar, settings, Alexa, PDF reader, Spotify, email, contacts, code checker, morning review, monthly reports.

Tier 2 - Code 4201 (D2L + Partner Wizard)

D2L ticker and partner shift scheduling wizard only.

Tier 3 - Code 1010 (D2L Ticker Only)

D2L announcement ticker only.

Passwords in env vars. Sessions in PostgreSQL via express-session.

3. File Structure

Root

package.json, tsconfig.json, vite.config.ts, drizzle.config.ts, tailwind.config.ts

server/

index.ts - Express entry
routes.ts - All API routes (~20,000+ lines)
storage.ts - DB CRUD (IStorage)
gmail.ts - Gmail OAuth
secondGoogleAccount.ts - Partner Google account
googleCalendar.ts / outlookCalendar.ts - Calendar integrations
onedrive.ts - OneDrive sync
vite.ts - Vite middleware

client/src/

pages/dashboard.tsx - Main dashboard (~18,700+ lines)
components/CourseDetailDialog.tsx, ui/ (shadcn)

shared/

schema.ts - Drizzle tables + Zod schemas

4. Database Schema

- tasks - Homework tracking
- files - PDFs/docs with TTS metadata
- weather_records - Hourly weather
- contacts - Contact entries
- notepad_notes - Notes
- app_state - Settings KV store
- session - Express sessions
- monthly_reports - Academic summaries
- partner_shifts - CRCU shifts
- d2l_announcements - D2L cache

5. All API Routes

Tasks

GET/POST/PATCH/DELETE /api/tasks
POST /api/tasks/batch-delete

Weather

GET /api/weather

Spotify

GET /api/spotify/now-playing, playlists
POST play, pause, next, previous, volume, shuffle, repeat

Home Assistant

GET /api/ha/status, entities, services
POST /api/ha/command, /api/speakers/stop-all

Webhooks

POST cat-lights, cat-lights-confirm, cat-shower-button
POST cat-wash-stop, cat-volume, cat-knob-press
POST voice-command, kitchen-volume, play-urgent-pdf

Files

GET /api/files, POST sync, PATCH :id, GET tts-audio

Other

POST publish-guide, contacts CRUD, notepad CRUD
POST email/search, GET d2l/announcements
POST morning-review/sync-all, monthly-reports/generate
POST code-checker/analyze, GET partner-shifts

6. Frontend Features

- Top Pill Bar - Frosted glass toolbar
- Calendar Grid - Weekly with resizable columns
- Day Detail View - 24-hour AM/PM layout
- Homework Panel - Left sidebar, grouped tasks
- Module Boxes - Course reading display
- Countdown Bars - Floating deadline indicators
- Weather Widget - Current + 7-day forecast
- News Ticker - Scrolling D2L + RSS

7. Dashboard Layout

Blink Settings

- buttonSpacing, tallPillButtonSpacing, countdownBarHeight, todayColumnBlink, headerBar

Course Colors

CPPA122: #47B045 CFNF400: #FA67B3 CASL101: #6366f1
CECN210: #34D399 CPHL110: #60A5FA CHIS105: #F87171
CPPA235: #CD853F Chang blue: #004C9C

Module Boxes

- White bg z-index 44, row z-index 45, solid opaque, time in white

8. Top Pill Bar - Undocking

Frosted glass toolbar supporting dock/undock and free-form drag.

Grab Handle

- 3x3 dot grid on LEFT side, vertically centered
- White dots when docked, BLACK dots when undocked
- Dragging undocks pill for free-form positioning

Redock Button

- Up-arrow below the chevron-up arrow
- Always visible - greyed out (25% opacity) when docked
- Black and clickable when undocked
- Snaps pill back to docked position

State

```
topPillUndocked/Ref, topPillPos/Ref, topPillDragRef  
closeTopPill guards against firing when undocked
```

9. Calendar & Day View

Calendar Grid

- Week: Sun-Sat, School week: Sat-Fri
- Time 6AM-midnight, module column toggleable
- Today column: black left border, blinking hour
- Overlapping tasks expand to full width on hover (z-index 48)

Day Detail View

- 24-hour AM/PM, ROW_PCT = 100/12
- Add button: pill style, gradient glass, Avenir font

10. Weather

- Open-Meteo API, no key. 7-day forecast.
- Upcoming days greyed until Saturday
- BUG: 0/0 Snow when dayForecast undefined

11. Homework Panel

- Due Today, This Week, Upcoming, Overdue groups
- Types: module, reading, essay, discussion, poll, quiz, exam, project, reminder, meeting, scholarship, class, other
- Swipeable rows, priority colors, calendar invites, Alexa reminders

NEVER modify CASL data directly in DB.

BUG: POST /api/tasks toISOString error - use SQL workaround.

12. News Ticker

- D2L from Gmail (NotificationEmail@toronto-mu.brightspace.com)
- Shimmer effect for urgent items, pushed to HA sensors

13. Spotify

- Web API with OAuth, play/pause/next/prev/volume/shuffle/repeat
- Playlist browsing, album art display

14. Alexa Speakers

- Dialog with individual + STOP ALL controls
EVERYWHERE: `media_player.byhome`
BATHROOM/KITCHEN: `bathroom_echo`, `kitchen_echo`
CAT: `echo_cat_left/right/middle`

15. Calendar System

- Google Calendar: TMU events, server/googleCalendar.ts
- Outlook Calendar: server/outlookCalendar.ts
- Both use Replit connectors (rewrite for Pi)

16. CRCU Shifts

- Partner shifts, Tier 2 wizard, second Google account

17. Morning Review

- Daily summary dialog on first login

18. Monthly Reports

- Auto-generated academic summaries

19. OneDrive & PDF Reader

- File sync, PDF viewer, chunk-based TTS with fallback chain

20. Email

- Gmail: homeworkbryn@gmail.com (sending, D2L parsing)
- Outlook: bryn.kai-hendricks@outlook.com (receiving)
- Second Google account for partner features

21-23. Notepad, Code Checker, Contacts

- Notepad: CRUD via /api/notepad/notes
- Code Checker: OpenAI analysis, email results
- Contacts: categories, favorites, key contacts dialog

24. Countdown Bars

- Floating on calendar, color-coded urgency, sticky scroll, Xd Yh format

25. Settings

All in app_state under blinkSettings. Includes spacing, widths, colors, animations.

26. Git & Pi Migration

Remote: <https://github.com/Reachushere/Home-View.git>

Pi: SSH -> Node 20 + PostgreSQL -> Clone -> npm install -> .env -> db:push -> build -> systemd service. See Self-Hosting Guide in original PDF.

27. Environment Variables

DATABASE_URL, SITE_PASSWORD (5747/4201/1010), SESSION_SECRET
GOOGLE_CLIENT_ID/SECRET/REFRESH_TOKEN
SPOTIFY_CLIENT_ID/SECRET/REFRESH_TOKEN
HOME_ASSISTANT_TOKEN, DEPLOYED_APP_URL
OPENAI_API_KEY, GOOGLE_SECOND_ACCOUNT_*

28. Deployment

- Replit: npm run dev, published at home-view--bkh416.replit.app
- Pi: npm run build, systemd + Nginx, update HA webhook URLs

29. Cat Washroom Study System

Automated TTS study reading when bathroom lights turn on.

Flow A: Lights ON

Webhook -> select urgent PDF -> TTS prompt -> confirm -> play

Flow B: Lights OFF

Stop speakers, save position

Flow C: Confirmed Play

Sequential chunks, progress tracked

Flow D: Shower Button

Skip confirmation, play immediately

Flow E: Volume

Adjust cat WR speaker volume

Flow F: Knob Press

Master STOP

Flow G: Toothbrush

Auto-pause on sensor

Flow H: Voice

pause/resume/stop/skip/restart/reset

Flow I: Urgent PDF

Play most urgent unlistened PDF

TTS Fallback

1. OpenAI -> 2. Edge TTS -> 3. espeak-ng

30. HA YAML Reference

- REST commands for each webhook in configuration.yaml
- 6 core automations for Cat Washroom triggers
- input_boolean.cat_washroom_reading_active
- Update URLs to Pi address when migrating

31. Troubleshooting

- App won't start: check DATABASE_URL, db:push, port 5000
- OAuth expired: redo flow, publish consent screen
- Weather 0/0: guard undefined dayForecast
- Tasks toISOString: use direct SQL
- D2L blocked: enable Gmail API for project 690968796215
- Pi connectors: rewrite 4 files for direct OAuth